

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B.Tech Examination (IT/CE/EC/ME/CL/EE)

November 2013

CE101: Fundamentals of Computing & Programming

Date: 30.11.2013, Saturday

Time: 10:00 am to 01:00 pm

Maximum Marks:70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions:

- | | |
|---|----|
| (a) What is the output of <code>printf ("%d")?</code> | 02 |
| (b) What is modular programming? Describe it with necessary figures. | 03 |
| (c) Explain the basic structure and steps of execution of a C program. | 03 |
| (d) State True or False: | 03 |
| 1. "If" statements are also referred to as Conditional Execution. | |
| 2. A conditional expression is a question that will evaluate to either true or false. | |
| 3. Comments are instructions that cause some action to take place. | |

Q-2 Write down the following C Programs:

12

- | | |
|---|--|
| (a) Write a program to find the number of and sum of all integers greater than 100 and less than 200 that are divisible by 7. | |
| (b) Write a program to copy a given string s1 into another string s2 without <i>string.h</i> file. | |

OR

Q-2

- (a) Admission to a professional course is subject to the following conditions:

- | |
|---|
| (a) Marks in mathematics ≥ 60 |
| (b) Marks in physics ≥ 50 |
| (c) Marks in chemistry ≥ 40 |
| (d) Total in all three subjects ≥ 200 or Total in mathematics and physics ≥ 150 |

Given the marks in the three subjects, write a program to process the applications to list the eligible candidates.

- (b) Write a program using pointers to exchange the value of two variables.

Q-3 Attempt Any Three.

12

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|--|
| (a) What is an infinite loop? Write infinite loop using while, do...while and for loop constructs. |
| (b) How do variables and symbolic names differ? |
| (c) In what ways does a switch statement differ from if statement? |
| (d) Describe the limitations of using <i>getchar</i> and <i>scanf</i> functions for reading strings. |
| (e) What is a pointer? How is a pointer initialized? |

SECTION-II

Q-4 Do as directed:

(a) Choose the appropriate option:

02

1. A program is:

- a) Set of implementations
- b) A list of instructions (actions) to accomplish a task.
- c) What you watch on TV
- d) None of the above

2. "If it is later than 4:50 pm, class is over", is an example of a:

- a) condition
- b) decision
- c) loop
- d) method
- e) None of the above

(b) State True/False:

09

1. A float is 4 bytes wide, whereas a double is 8 bytes wide.
2. Functions cannot return more than one value at a time.
3. A preprocessor directive is a message from programmer to the preprocessor.
4. A header file contains macros, structure declaration and function prototypes.
5. float, double, long double are the different types of real data type in C.
6. For the following statements will arr[3] and ptr[3] fetch the same character?
`char arr[] = "IndiaBIX";`
`char *ptr = "IndiaBIX";`
7. In the expression `a=b=5` the order of Assignment is NOT decided by Associativity of operators.
8. Bitwise & and | are unary operators.
9. It is not necessary that for the string functions to work safely the strings must be terminated with '\0'.

Q-5 Write down the following programs:

12

(a) Define a structure called cricket that will describe the following information:

Player name
Team name
Batting average

Using cricket, declare an array player with 5 elements and write a program to read the information about all the 5 players and print a team-wise list containing names of players with their batting average.

(b) Write a program that uses the functions ftell and fseek.

OR

Q-5

(a) Define a structure that can describe a hotel. It should have members that include the name, address, grade, average room charge, and number of rooms.

Write a functions to perform the following operations:

- (a) To print out hotels of a given grade in order of charges.
- (b) To print out hotels with room charges less than a given value.

(b) A File named DATA contains a series of integer numbers. Code a program to read these numbers and then write all odd numbers to a file to be called ODD and all even numbers to a file to be called EVEN.

Q-6 Attempt Any Three.

12

- (a) When a program is terminated, all the files used by it are automatically closed. Why is it then necessary to close a file during execution of the program?
- (b) What is the use of sizeof? Give an example for it.
- (c) main is a user-defined function. How does it differ from other user-defined functions?
- (d) Distinguish between the following:
 - (a) Actual and formal arguments
 - (b) Structure and Union
- (e) Explain the need for array variables.
